P1-C Group 3: analytical multilayer script, validation report

EM wave penetration in layered tissue (2.45 GHz team). Prepared by Yusuf Uzun, 2026-11-02. Evidence for Checkpoint 1. Scope: the Python script, the Feko plane-wave results (section 7) and the Feko dipole results (section 8).

1. Checks against the pass criteria of the workbook

Check	Reference	Pass criterion	Result	Pass?
Script: single interface	Fresnel formula, air/muscle, 4 frequencies	4 significant digits	equal at all 4 frequencies; max |Γ| difference 2.2e-16	PASS
Script: λ/4 matching layer	Textbook, lossless layer, η1 = √(η0·η2)	|Γ| < 1e-6	|Γ| = 2.2e-16	PASS
Penetration depth in muscle (script vs Week 1 table)	Week 1 slide: 2.23 cm at 2.45 GHz	within 2 %	δ = 22.33 mm (0.13 % from 22.3 mm)	PASS
Script: power conservation	1 − |Γ|² = Σ P_abs/S_inc + P_trans/S_inc	small residual	max error 7.2e-10	PASS
Tissue media vs Week 1 table	Week 1 slide 17 (ε′, σ at 4 frequencies)	within 1 %	worst 0.90 % (fat eps_r at 5800 MHz)	PASS
Tissue media vs course media pack	Course files Muscle / Skin_dry / Fat_not_infiltrated (216 points each, 1 MHz to 20 GHz)	within 1 %	worst 0.0031 % at the file's own frequencies	PASS
Feko vs script: |Γ|	Feko PBC cells: 60/90 mm muscle slab, skin/fat/muscle stack	within 3 %	worst 0.24 % over 5 runs	PASS
Feko vs script: absorbed power per layer	power loss per medium / incident power on the cell	within 3 %	worst 1.52 % (relative, per layer)	PASS
Feko vs script: |E| along the centre line	near-field line, interface points skipped	within 3 %	largest layer mean 1.27 %	PASS
Power balance (plane-wave cell)	power loss vs 1 − |R|² − |T|²	error < 5 %	worst 0.83 %	PASS
Feko δ in muscle slab (Week 1 window)	analytic δ = 22.33 mm	within 2 %	60 mm: +1.8 %, 90 mm: -0.25 %	PASS
Mesh convergence (Feko)	last refinement vs previous: |Γ| (and absorbed power in section 7)	within 2 %	worst 0.055 %	PASS
Power balance (dipole case)	P_acc = P_rad + P_abs (far-field integral)	error < 5 %	0.05 % (coarse mesh), 0.21 % (fine mesh)	PASS

2. Test 1: air/muscle interface vs Fresnel

Muscle half-space, Gabriel properties at each frequency, normal incidence. Reference Γ = (1 − n)/(1 + n) with n = √ε*. Compared as complex numbers (difference below 1e-9).

f (MHz)	script |Γ|	Fresnel |Γ|	1 − |Γ|² (power entering)	4 digits
403	0.790354	0.790354	37.53 %	PASS
915	0.770809	0.770809	40.59 %	PASS
2450	0.762460	0.762460	41.87 %	PASS
5800	0.756698	0.756698	42.74 %	PASS

3. Test 2: lossless quarter-wave layer

Layer ε1 = 3 (λ1/4 thick) on a lossless ε2 = 9 half-space, η1 = √(η0·η2). At the design frequency |Γ| = 2.2e-16. As a counter-check, at twice the frequency the layer is λ/2 thick and the stack must behave like the bare interface: |Γ| = 0.500000 versus 0.500000 for the bare interface.

4. Power conservation, skin 1.5 mm / fat 8 mm / muscle half-space

Absorbed power per layer is the drop of the Poynting flux across the layer. It is cross-checked by integrating ½σ|E|² over the layer (last column).

f (MHz)	|Γ|	1 − |Γ|²	Σ P_abs/S_inc	P_trans/S_inc	conservation error	Poynting vs ∫½σ|E|²
403	0.7665	0.41250	0.41250	0.0e+00	1.4e-10	3.3e-07
915	0.6515	0.57561	0.57561	0.0e+00	2.5e-10	3.3e-07
2450	0.5231	0.72636	0.72636	0.0e+00	6.3e-10	3.3e-07
5800	0.8669	0.24851	0.24851	0.0e+00	7.2e-10	3.3e-07

5. Penetration depth and the finite-slab bias

Muscle half-space at 2.45 GHz: analytic δ = 22.330 mm, fit of ln|E| (Week 1 procedure, 5 mm skipped at each face) = 22.330 mm (+0.00 %). For a 60 mm muscle slab backed by air, as in a Feko model, the back-face reflection (|Γ| about 0.76 for muscle/air) adds a ripple: the same fit on the exact field gives 22.80 mm (+2.12 %), which is close to the 2 % limit. Skipping 1.5 δ at the bottom of the slab gives 22.43 mm (+0.47 %). A Feko δ check should therefore use both windows.

6. Unit tests (pytest)

Result: 10 passed in 0.52s. Environment: Python 3.13.16, numpy 2.5.3, matplotlib 3.11.2.

Test	Result
test_fresnel_4_digits	PASSED
test_fresnel_matches_week1_slide	PASSED
test_quarter_wave_match	PASSED
test_power_conservation	PASSED
test_field_continuity	PASSED
test_gabriel_vs_week1_table	PASSED
test_penetration_depth_fit	PASSED
test_penetration_depth_fit_finite_slab	PASSED
test_dipole_reference	PASSED
test_parse_feko_out_fixture	PASSED

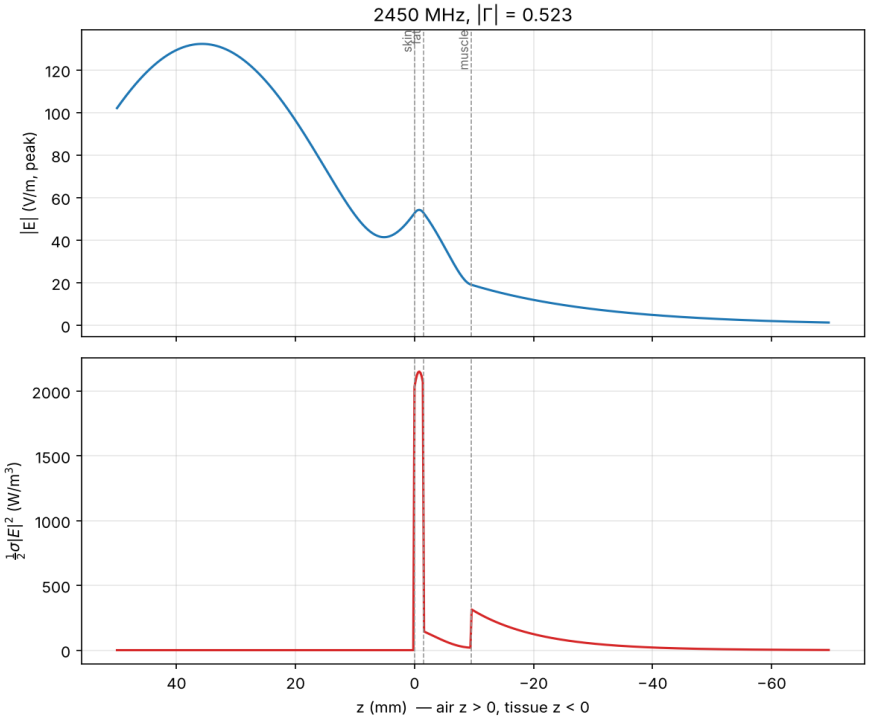


Figure 1. |E|(z) and ½σ|E|²(z) at 2.45 GHz for skin 1.5 / fat 8 / muscle half-space (10 W/m² incident; air on the left).

7. Feko results against the script (plane wave, periodic 10 x 10 mm cell, FEM)

Feko 2022, 2.45 GHz, |E0| = 86.8 V/m peak (10 W/m²), second-order FEM tetrahedra, periodic boundary cell, media from the course pack (checked in the .out files: eps_r and sigma equal the model values to 0.04 %). The script solves the same stack with the same slab backed by air. Absorbed power is the Feko power loss per medium divided by the power incident on the cell.

Run	Case	Mesh (mm)	Tets	DOF	Mem GB	Time s	|Γ| Feko	|Γ| script	diff
B2	muscle slab T = 60 mm	1.11	33647	205256	1.52	9	0.75982	0.76165	-0.24 %
B3	muscle slab T = 60 mm	0.83	81851	503766	2.65	63	0.76023	0.76165	-0.19 %
B4	muscle slab T = 90 mm	1.11	50917	310837	1.77	44	0.76238	0.76246	-0.01 %
C1	skin / fat / muscle 60 mm	0.5 / 1.0 / 1.11	52607	321869	1.92	34	0.52257	0.52229	+0.05 %
C2	skin / fat / muscle 60 mm	0.4 / 0.8 / 0.83	116033	715365	3.34	91	0.52265	0.52229	+0.07 %

Run	Layer	Feko (% of incident)	script (%)	diff	|E| mean diff	|E| max diff
B2	Muscle	42.541	41.906	+1.52 %	1.27 %	10.46 %
B3	Muscle	42.403	41.906	+1.18 %	1.00 %	8.23 %
B4	Muscle	41.868	41.859	+0.02 %	0.12 %	0.88 %
C1	skin	31.620	31.670	-0.16 %	0.09 %	0.09 %
C1	fat	5.508	5.515	-0.13 %	0.08 %	0.09 %
C1	Muscle	35.416	35.466	-0.14 %	0.18 %	0.59 %
C2	skin	31.607	31.670	-0.20 %	0.09 %	0.10 %
C2	fat	5.505	5.515	-0.18 %	0.10 %	0.11 %
C2	Muscle	35.392	35.466	-0.21 %	0.21 %	0.73 %

Convergence and truncation. Model B, 1.11 to 0.83 mm: |Γ| changes by +0.055 %. Stack, mesh edges reduced by 1.25 to 1.34 times (52 607 to 116 033 tetrahedra): |Γ| +0.015 %, absorbed power skin -0.04 %, fat -0.05 %, Muscle -0.07 %. Truncation: T = 90 mm gives |Γ| = 0.76238 (script 0.76246), T = 60 mm differs from it by -0.34 %. The 60 mm slab is enough for |Γ| and the near-surface field. Deeper in the muscle the back-face echo changes |E| (the script predicts the same change), so the penetration-depth fit uses the window that skips 1.5 δ at the back.

Open point in the 60 mm muscle slab (runs B2, B3). There the Feko absorbed power is 1.2 to 1.5 % above the script, the energy-balance gap is 0.7 to 0.8 %, and |E| differs by about 1 % on average and by up to 8 to 10 % near the minimum close to the back face. With the 90 mm slab (B4) the absorbed power agrees to 0.02 % and the balance gap is 0.01 %. The stack, which has the same 60 mm of muscle, agrees to 0.2 %. The cause of the 60 mm muscle-slab offset is not identified yet; all criteria are still met (3 % and 5 %), but the offset has to be explained before the 60 mm slab is used for anything finer than that.

Penetration depth. With the Week 1 window (5 mm off each face) Feko gives δ = 22.73 mm (+1.8 %) in the muscle-only 60 mm slab at both mesh sizes and 22.27 mm (−0.25 %) in the 90 mm slab, so the 2 % criterion is met. For the muscle layer of the stack the same window gives +2.1 % on the Feko field and +2.2 % on the exact script field, so that excess is the finite-slab bias of the window and not a Feko error (section 5). Skipping 1.5 δ at the back gives +0.1 to +0.4 % in all three cases.

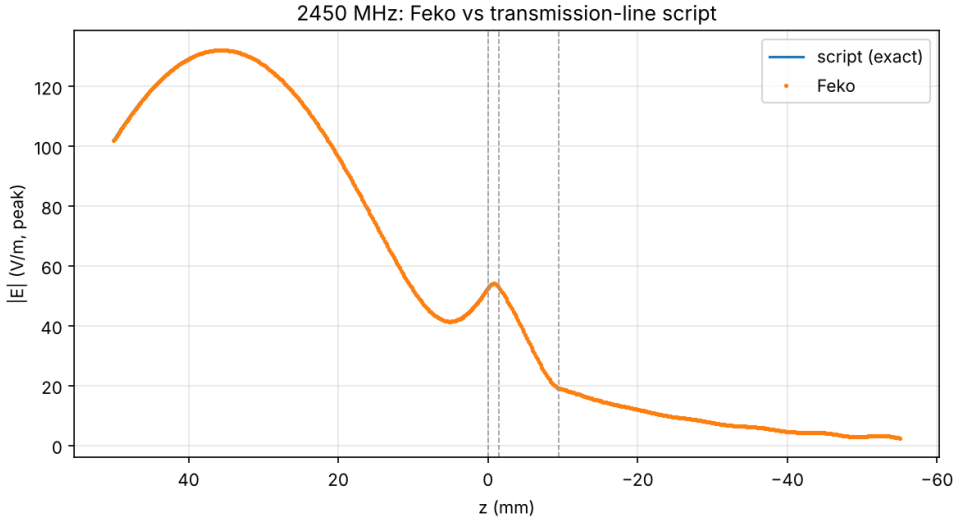


Figure 2. Stack, finest mesh: Feko |E| along the centre line against the script.

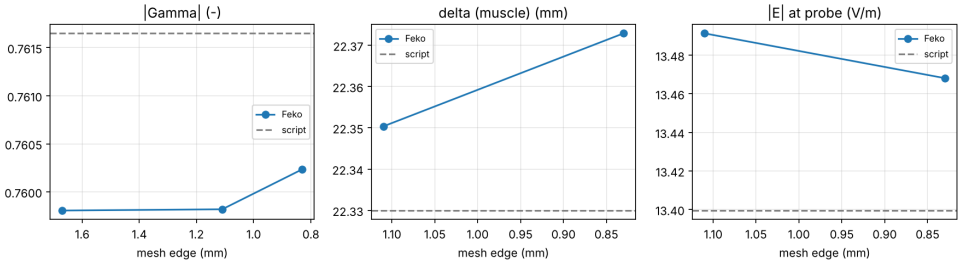


Figure 3. Model B (60 mm muscle slab): quantities against mesh size.

8. Dipole 10 mm above the slab (Feko, FEM slab with MoM wire)

A half-wave wire dipole (radius 0.1 mm, L = 58.3 mm = 0.476 λ0, tuned in free space to X = 0 at 2.45 GHz) lies parallel to a 60 x 60 mm slab of skin 1.5 mm, fat 8 mm and muscle 56 mm, 10 mm above its surface; it is fed in the centre with 1 V (50 Ω reference). An electric symmetry plane at x = 0 and a magnetic one at y = 0 reduce the memory from 8.9 GB (the run without symmetry aborted) to 6.0 GB (189 220 tetrahedra, 300 796 unknowns, 143 s per frequency). The radiated power is the far-field integral, independent of the loss integrals.

Quantity	Free space	Above the slab
Resonance (X = 0)	2.4516 GHz	2.4844 GHz
Resistance at resonance	71.8 Ω	65.8 Ω
Best |S11| (50 Ω)	-15.0 dB (sample)	-16.6 dB (sample)
−10 dB band	2.359 to 2.532 GHz (173 MHz)	2.377 to 2.585 GHz (208 MHz)
Z_in at 2.45 GHz	71.6 − j0.6 Ω	64.5 − j11.6 Ω
|S11| at 2.45 GHz	−15.0 dB	-15.8 dB
Accepted power (1 V)	6.98 mW	7.51 mW
Radiated power (far field)	6.98 mW	2.05 mW
Absorbed power (skin / fat / muscle)	0	2.13 / 0.50 / 2.84 mW
Radiation efficiency	100 %	27.3 %

Power balance. P_acc = P_rad + P_abs holds within 0.05 % (coarse mesh, 105 752 tetrahedra) and 0.21 % (fine mesh); the criterion is 5 %. Mesh. Refining from 105 752 to 189 220 tetrahedra changes Z_in by +0.02 % (R) and -0.15 % (X) and the loss per layer by at most 0.41 %. Layer split. Of the absorbed power 38.9 % is in skin, 9.2 % in fat and 51.9 % in muscle (plane wave: 43.6 / 7.6 / 48.8 %). Detuning. The slab moves the resonance by +32.8 MHz (+1.34 %) and widens the −10 dB band from 173 to 208 MHz (frequency sweep 2.30 to 2.60 GHz).

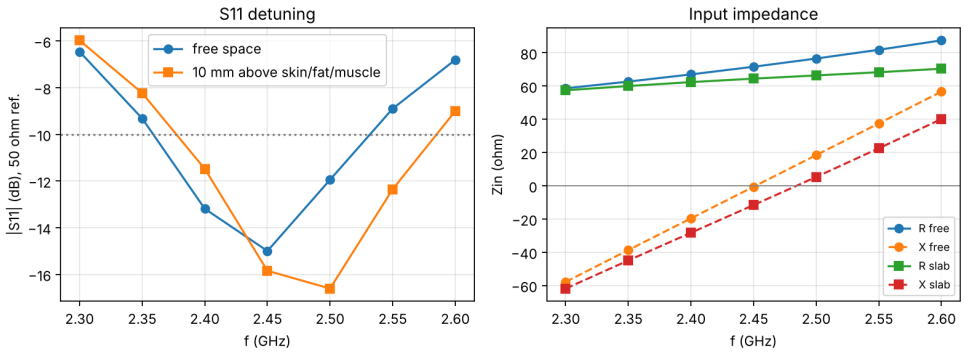


Figure 4. |S11| and input reactance of the dipole in free space and above the slab.

Free-space impedance and wire radius. At exactly 0.5 λ0 (61.18 mm) the resistance depends on the wire radius: 91.5, 84.1 and 80.4 Ω for radius 0.5, 0.1 and 0.02 mm (X = 46.2, 45.6, 44.2 Ω), against 73.1 + j42.5 Ω of the induced-EMF reference for an infinitely thin wire. Extrapolating the trend gives 75 to 77 Ω, so about 5 % remains. At the tuned length the resistance is 72 Ω.

9. Limitations and open items

• Feko plane-wave cases are in section 7 and the dipole in section 8. The 60 mm muscle-only slab shows an absorbed-power offset of 1.2 to 1.5 % that is not explained (within the 3 % criterion).
• Plane-wave runs use a single frequency (2.45 GHz) and normal incidence; the |E| line stops 55 mm below the surface, so the last 15 mm of the 69.5 mm stack are checked through |Γ| and the absorbed power only. The dipole was run at 2.30 to 2.60 GHz; 403, 915 and 5800 MHz are script-only.
• The muscle thickness (60 mm, 56 mm for the dipole) follows the 2.5 δ rule of the specification instead of the 50 mm of the stack; in the script 50 mm changes |Γ| by 0.5 % and the layer powers by less than 1 %.

• The tissue properties are computed from the published Gabriel (1996) four-term Cole-Cole parameters. They agree with the course media pack files to 0.0031 % at the file frequencies, and with the Week 1 table to 0.90 %.
• The tests check the script against analytical references and the lecture numbers, so they cannot reveal a convention mismatch with Feko; that is what the Feko vs script comparison is for.